Case #1:

Raw data

Happy Supplies
Parts Warehouse

Customer Name:	Jeff Peterson	Date:	7/1/2024
Customer Number:	H-5547 587	Time:	10:30am
		Employee:	D. Harrison
Customer Type:	Consumer		

Part Number	Name	Type	Cage Code	Quantity Ordered	Unit Price
10654	Float Control	Plumbing	G413	4	12
10456	Modulator	Electrical	H433	3	7
10776	Hose Assembly	Plumbing	G413	7	9
10657	Float Assembly	Plumbing	G413	5	10

Assumptions:
- An order is considered unique based on the customer ID (customerNumber), date, time.
- A part is identified by the partNumber. Attributes: partName, partType, cageCode, unitPrice depend on partNumber
- QuantityOrdered depends on each order transaction (particular order)
- We assume that one customer can place many orders

1NF

customerNumber	customerName	customerType	date	time	employee	partNumber	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

1NF is established from the flattened raw form:
Assumed composite key: customerNumber, date, time, partNumber
Single, atomic value in each field + composite key to uniquely identify row

2NF

customer

customerNumber (PK)	customerName	customerType
HG54587	Jeff Peterson	Consumer
HG54587	Jeff Peterson	Consumer
HG54587	Jeff Peterson	Consumer
HG54587	Jeff Peterson	Consumer

part

partNumber (PK)	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order

customerNumber (PK)	date (PK)	time (PK)	employee
HG54587	7/1/2024	10:30 am	D. Harrison
HG54587	7/1/2024	10:30 am	D. Harrison
HG54587	7/1/2024	10:30 am	D. Harrison
HG54587	7/1/2024	10:30 am	D. Harrison

order_details

customerNumber (PK)	date (PK)	time (PK)	partNumber (PK)	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

Numerous *partial dependencies* arise in the 1NF, where some attributes depend only on the part of the defined composite key, like customerName and customerType depending on customerNumber. quantityOrdered, partName, partType, cageCode depend on partName. Due to this, we separate the attributes with these partial dependencies into their own relations.

customer primary key: customerNumber
order primary key: customerNumber, date, time
part primary key: partNumber
order_details primary key: customerNumber, date, time, partNumber

3NF

customer

customerNumber (PK)	customerName	customerType
HG54587	Jeff Peterson	Consumer
HG54587	Jeff Peterson	Consumer
HG54587	Jeff Peterson	Consumer
HG54587	Jeff Peterson	Consumer

part

partNumber (PK)	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order

customerNumber (PK, FK to customer)	date (PK)	time (PK)	employee
HG54587	7/1/2024	10:30 am	D. Harrison
HG54587	7/1/2024	10:30 am	D. Harrison
HG54587	7/1/2024	10:30 am	D. Harrison
HG54587	7/1/2024	10:30 am	D. Harrison

order_details

customerNumber (PK, FK to order)	date (PK, FK to order)	time (PK, FK to order)	partNumber (PK, FK to part)	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

We build from the 2NF structure, and verify if we have any transitive dependencies, which we do not have. Therefore, 2NF becomes our 3NF with defined Foreign Keys.

customer primary /foreign key: customerNumber (PK)
order primary/foreign key: customerNumber (PK, FK to customer), date(PK), time(PK)
part primary/foreign key: partNumber (PK)
order_details primary/foreign key: customerNumber (PK, FK to order), date (PK, FK to order), time (PK, FK to order), partNumber (PK, FK to part)

Case #2:

Raw data

staffNo	therapistName	patNo	patName	appointment date	appointment time	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Assumptions:
- Therapist's unique ID is staffNo
- Appointment is identified through patNo, appointment date/time
- Patient can't be in more than one place/branch
- Therapists can work in different branches but can only work in one branch for a day (determined by appointment date and staffNo)

0NF

staffNo	therapistName	patNo	patName	appointment date time	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022 10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022 12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022 10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022 14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022 16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	09/15/2022 18:00	B13

0NF is established from the raw form:
Multi-valued cells under appointment date time

1NF

staffNo	therapistName	patNo	patName	appointmentDate	appointmentTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022 10:00	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022 12:00	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022 10:00	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022 14:00	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022 16:30	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	09/15/2022 18:00	18:00	B13

1NF is established from the raw form:
Assumed composite key: patNo, appointmentDate, appointmentTime
We change the multi valued cells under appointment date time by separating the columns into appointmentDate and appointmentTime.

2NF

therapist

staffNo (PK)	therapistName
S1011	Fred Smith
S1011	Fred Smith
S1024	Heidi Pierce
S1024	Heidi Pierce
S1032	Richard Levin
S1032	Richard Levin

patient

patNo (PK)	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P108	Andy McKee
P105	Jill Baker
P110	Jimmy Winter

therapist_branch

staffNo (PK)	appointmentDate (PK)	branchNo
S1011	9/12/2022	D. Harrison
S1011	9/12/2022	D. Harrison
S1024	9/12/2022	D. Harrison
S1024	9/14/2022	D. Harrison
S1032	9/14/2022	
S1032	09/15/2022	

appointment

patNo (PK)	appointmentDate (PK)	appointmentTime (PK)	staffNo
P100	9/12/2022	10:00	S1011
P105	9/12/2022	12:00	S1011
P108	9/12/2022	10:00	S1024
P108	9/14/2022	14:00	S1024
P105	9/14/2022	16:30	S1032
P110	09/15/2022	18:00	S1032

Some *partial dependencies* arise in the 1NF, as patName depends only on patNo, therapistName only depends on staffNo, branchNo is dependent on staffNo and appointmentDate because therapists are limited to one branch per day.

therapist primary key: staffNo
therapist_branch primary key: staffNo, appointmentDate
appointment primary key: patNo, appointmentDate, appointmentTime
patient primary key: patNo

3NF

therapist

staffNo (PK)	therapistName
S1011	Fred Smith
S1011	Fred Smith
S1024	Heidi Pierce
S1024	Heidi Pierce
S1032	Richard Levin
S1032	Richard Levin

patient

patNo (PK)	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P108	Andy McKee
P105	Jill Baker
P110	Jimmy Winter

therapist_branch

staffNo (PK, FK to therapist)	appointmentDate (PK)	branchNo
S1011	9/12/2022	D. Harrison
S1011	9/12/2022	D. Harrison
S1024	9/12/2022	D. Harrison
S1024	9/14/2022	D. Harrison
S1032	9/14/2022	
S1032	09/15/2022	

appointment

patNo (PK, FK to patient)	appointmentDate (PK, FK to therapist_branch)	appointmentTime (PK)	staffNo (FK to therapist_branch)
P100	9/12/2022	10:00	S1011
P105	9/12/2022	12:00	S1011
P108	9/12/2022	10:00	S1024
P108	9/14/2022	14:00	S1024
P105	9/14/2022	16:30	S1032
P110	09/15/2022	18:00	S1032

We build from the 2NF structure, and verify if we have any transitive dependencies, which we do not have. Therefore, 2NF becomes our 3NF with defined Foreign Keys.

therapist primary/foreign key: staffNo (PK)
therapist_branch primary/foreign key: staffNo (PK, FK to therapist), appointmentDate (PK)
appointment primary/foreign key: patNo(PK, FK to patient), appointmentDate (PK, FK to therapist_branch), appointmentTime(PK), staffNo (FK to therapist_branch)
patient primary/foreign key: patNo (PK)

Case #3:

Raw data

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Assumptions:
- Employee's unique ID is eNo and eName is dependent on this ID
- Hours worked depends on eNo (employee ID) and the contactNo (type of contract)
- Event's unique ID is eventNo and eventLoc is dependent on this ID
- Contract's unique ID is contractNo and applies to only one event

1NF

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

1NF is established from the raw form:
assumed composite primary key: eNo, contractNo
Single, atomic values in each field + composite primary key

2NF

employee

eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T
1135	Smith J
1135	Smith J

contract

contractNo (PK)	eventNo	eventLoc
C1024	H25	Queens
C1024	H25	Queens
C1025	H4	Yonkers
C1025	H4	Yonkers
C1026	H25	Queens

assignment

eNo (PK)	contractNo (PK)	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

Some *partial dependencies* arise in the 1NF, as eName only depends on eNo, eventLoc and eventNo only depends on contractNo, and hours depends on contractNo and eNo.

employee primary key: eNo
contract primary key: contractNo
assignment primary key: eNo, contractNo

3NF

event

eventNo	eventLoc
H25	Queens
H25	Queens
H4	Yonkers
H4	Yonkers
H25	Queens

employee

eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T
1135	Smith J
1135	Smith J

contract

contractNo (PK)	eventNo (FK to event)
C1024	H25
C1024	H25
C1025	H4
C1025	H4
C1026	H25

assignment

eNo (PK, FK to employee)	contractNo (PK, FK to contract)	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

We build from the 2NF structure, and verify if we have any transitive dependencies, which we *do* have. Within the contract relation, a transitive dependency emerges, as contractNo determines eventNo and eventNo determines eventLoc. We thus decompress contract into two relations: contract and event.

event primary/foreign key: eventNo (PK)
contract primary/foreign key: contractNo (PK), eventNo (FK to event)
assignment primary/foreign key: eNo (PK, FK to employee), contractNo(PK, FK to contract)
employee primary/foreign key: eNo (PK)